NATIONAL
Junior College

SH2 Preliminary Examination 2026

9173 H2 Geography Paper 2

Suggested Marking Guide

Disclaimer:
The Suggested Marking Guide should be treated with confidentiality, and should be used with discretion.

Notes on Marking

Marking Approaches

a. **Point marking** is used for marking all sub-parts of the structured questions, except for the 10-mark fieldwork evaluative sub-part.

   For point marking, the general rule is that each creditworthy response is awarded one mark. More detailed guidance is provided in the marking guide for each question.

b. **Generic level descriptors** are used for marking:

- 10-mark fieldwork evaluative sub-part structured question; and
- 20-mark essays.

Possible approaches are provided for questions marked using generic level descriptors. They provide guidance on how the questions could be approached. The possible approaches provided are **neither exhaustive nor should they be treated as model approach to questions**.

Generic Level Descriptors

Levels	Marks	Generic Level Descriptors for H2 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Note:
Assessment using generic level descriptors involves qualitative rather than quantitative evaluation. Judgements on the level to be awarded to an answer will be based on the principle of 'best fit' determined by the descriptors within each level.

As a general guideline, responses deemed to have fulfilled all the descriptors within a level may be awarded the top mark in that level.

Section A

Cluster 4: Fieldwork

Question	Answer	Marks
1	A group of five students in Singapore carried out fieldwork to investigate the attitudes and behaviours of adults aged 21–65 towards e-waste recycling and repair as environmentally sustainable practices. According to the National Environment Agency (NEA), e-waste is electrical and electronic equipment of any kind that has been discarded and includes anything powered by an electrical source, such as from a power socket or a battery. Their hypothesis was: *Younger adults aged 21 to 45 residing in Ang Mo Kio are more likely to participate in sustainable practices with respect to e-waste than those aged 46 to 65.* The students chose Ang Mo Kio as it is a mature residential estate with an ethnic mix that mirrors Singapore's national ethnic composition. They used systematic sampling, selecting every 30th person outside Ang Mo Kio Hub to complete a questionnaire, until they obtained a total of 50 respondents aged 21 to 65. They conducted their survey on a Tuesday and Wednesday morning during the March holidays. They obtained consent from the respondents and verified their ages before administering the questionnaire. Resource 1A is a map showing the location of Ang Mo Kio Hub. Resource 1B is a photograph of Ang Mo Kio Hub. Resource 2 shows selected demographic characteristics of Ang Mo Kio in comparison to Singapore. Resource 3 shows the questionnaire used by the students. Resource 4 shows responses to selected questions in the questionnaire.	
1(a)	**Explain <u>one</u> strength and <u>one</u> weakness of the students' hypothesis.** *Award 1 mark for each strength or weakness.* *Award a maximum of 1 additional mark for further development, where applicable.* *Award a maximum of 2 marks for explanation of <u>either</u> strengths <u>or</u> weaknesses only.* Possible responses include: Strength: • The hypothesis is **clearly defined** and **capable of being researched**, as it specifies **two distinct age groups** (21 – 45 and 46 – 65). o This allows the students to **operationalise the hypothesis** through a structured questionnaire and	**4**

Question	Answer	Marks
	draw clear comparisons between the two groups, making it straightforward to accept or reject the hypothesis based on the data collected. Weakness: • The hypothesis **assumes that age is the primary determinant** of sustainable behaviour towards e-waste, **without accounting for other variables such as income level, educational attainment or household type**, which may also significantly influence recycling and repair habits. o For example, a younger adult with low environmental awareness may be less likely to engage in sustainable practices than an older adult who has been exposed to community recycling programmes, meaning the **hypothesis oversimplifies the relationship between age and sustainable behaviour.** **AO1**	
1 (b)	**Using Resources 1A, 1B and 2, explain why the site chosen is suitable for the investigation.** *Award 1 mark for each explanation.* *Award a maximum of 1 additional mark for further development of each explanation, where applicable.* Possible responses include: • Ang Mo Kio Hub is a suitable site because it is **a major commercial and community hub** located in the heart of the estate (Resource 1), meaning it attracts **a high footfall of residents** from across the neighbourhood. o This increases the likelihood that the 50 respondents surveyed are actually **residents** of Ang Mo Kio rather than visitors from other areas. • Ang Mo Kio is a **mature residential estate** with its resident population spanning **a wide range of age groups**, including both the 21 – 45 and 46 – 65 age cohorts (Resource 2). • The fact that **Ang Mo Kio's ethnic distribution mirrors Singapore's national ethnic composition** (Resource 2) also adds to the site's suitability, as it **reduces ethnic bias in the sample**. **AO2**	**4**
1 (c)	**With reference to Resources 1A and 1B, suggest how the students could minimise the impact of their investigation.** *Award 1 mark for each suggestion* *Award a maximum of 1 additional mark for further development of each suggestion, where applicable.*	**4**

Question	Answer	Marks
	Possible responses include: • To minimise the impact of their investigation, the students should **position themselves away from the main entrances and exits** of the MRT station and bus interchange when approaching respondents / Students should **avoid positioning themselves directly outside the mall entrance** when administering their surveys. o This will minimise **obstructing the movement of commuters** who are not participating in the survey. / **blocking customer access** and **disrupting the businesses** to avoid having complaints from the staff or shop owners or management. • Students should also **seek prior permission from the management of Ang Mo Kio Hub** before conducting their survey on the premises, as **privately managed commercial spaces may have rules governing public surveys**. o Doing so would ensure that their **investigation is carried out with the knowledge and consent of relevant stakeholders**, **minimising any potential conflict** or disruption to the businesses. **AO1**	
1 (d)	**Using Resource 3, explain the strengths and weaknesses of the questionnaire.** *Award 1 mark for each explanation of strength / weakness.* *Award a maximum of 1 additional mark for further development of each explanation.* *Award maximum of 3 marks if only strengths or weaknesses are explained.* *Note: Only about the questionnaire, not about the time or sampling.* Possible responses include: Strengths • It has both open-ended and close-ended questions. This would allow the students to collect both **quantitative and qualitative data**. The open-ended question allows participants to efficiently answer the survey with the given choices. o The students can also **quickly tabulate the quantitative data** based on the pre-defined choices. / On the other hand, the open-ended questions, allow participants to **provide elaboration** and richer, more **nuanced qualitative data** that can reveal **deeper insights** into attitudes and motivations that pre-defined choices alone may not capture. • The design includes background information by asking for **age and residency** which ensures that the respondents fall within **target age groups of adults residing in Ang Mo Kio**.	**5**

Question	Answer	Marks
	Weaknesses: • One weakness of the questionnaire lies in its **question design**. Some questions are **poorly constructed** in ways that could compromise the reliability of the data collected. o Q4 asks how often respondents buy new electronics, but the **options do not include a "less than once every few years" or "never" category**, which may not capture respondents who rarely purchase electronics. The **options also assume a regular purchasing pattern** that may not reflect reality. • Another weakness is the **use of leading and assumptive language**. o The phrasing in Q10 "What would encourage you to recycle or repair your electronics?" assumes that the respondent is not already doing so, which may not be accurate for all respondents. A **more neutral phrasing** would acknowledge that some respondents may already be engaged in these behaviours. **AO2**	
1 (e)	**Explain how the students could improve the accuracy and reliability of the survey data collected.** *Award 1 mark for each explanation, to a maximum of 5 marks.* *Award a maximum of 1 additional mark for further development of each explanation.* *Award maximum of 3 marks if only accuracy / reliability is addressed.* *Note: **Accuracy** refers to how well the data reflects the true views of the target population (closeness to true reality / ground truth), i.e. whether the findings are free from bias and are genuinely representative. **Reliability** refers to the consistency and dependability of the data, i.e., whether the findings would be reproducible if the survey were repeated under similar conditions, and whether the sample size is sufficient to produce stable, trustworthy results.* Possible responses include: **Accuracy** • The students should **reconsider their sampling location**. Conducting the survey exclusively outside Ang Mo Kio Hub introduces **bias**. o The sample only **captures people who happen to visit that particular location, rather than a cross-section of all residents in the neighbourhood**. To improve representativeness, the students could conduct the survey at **different locations** across the neighbourhood, such as void decks and community centres. • Another improvement would be to **ensure a more balanced distribution of respondents across the two age groups**	**5**

Question	Answer	Marks
	being compared – 21 to 45 and 46 to 65 – **as well as across genders and ethnicities**. If one group is significantly over- or under-represented in the final sample, comparisons between the two age groups become less reliable. **Reliability** • The students should **reconsider their sampling timing**. Conducting the survey on Tuesday and Wednesday mornings during the March holidays introduces **bias.** o People present at a shopping hub on **weekday mornings** are likely to be retirees, homemakers or students having their school holidays, and are **unlikely to be representative of all working adults aged 21 to 65**. To improve representativeness, the students could conduct the survey **across different days and times of day**, including weekdays and weekends and evenings and weekends, to capture a wider range of respondents including working adults. **AO2**	
1 (f)	**Suggest how the data collected for Q1 in Resource 4 may be represented.** *Award 1 mark for each suggestion, to a maximum of 3 marks.* *Award 1 additional mark for further development of each suggestion.* *Note: There is <u>no need for actual plotting</u> of any graph for this question. While 1 mark may be awarded if the student is able to show how the data may be represented by plotting / sketching a graph, it is not necessary nor recommended due to the number of marks awarded. (1 mark for graph if plotted.)* *Do not accept compound / stacked bar graph as it does not allow easy visual comparison of the two age groups.* Possible responses include: • A **comparative / grouped bar graph** would work well as it allows for **easy visual comparison** across the two age groups for the four response options. o The **horizontal axis** would display the **four familiarity levels** (Very familiar, Somewhat familiar, Not very familiar, Not familiar at all) while the **vertical axis** would show the **number or percentage of respondents**. o For each familiarity level, **two bars would be placed side by side** — one representing the 21 to 45 age group and another representing the 46 to 65 age group, **differentiated by colour with a legend**.	**3**

Question	Answer	Marks
	• **Two pie charts** could be used for **side-by-side visual comparison** across the two age groups for the four response options. o **Each pie chart** would represent **one specific age group** and **the sectors within each chart represent the four familiarity levels** (Very familiar, Somewhat familiar, Not very familiar, Not familiar at all). o Use the **exact same colour palette for both charts.** o Arrange the four familiarity levels in the **exact same order in both charts** (using the Likert scale – Very familiar, Somewhat familiar, Not very familiar, Not familiar at all. AO2	
1 (g)	**In addition to the data collected by the students, suggest <u>other</u> types of data that they could have collected to help them in their investigation.** *Award 1 mark for each suggestion, to a maximum of 5 marks.* *Award a maximum of 1 additional mark for further development of each suggestion.* Possible responses include: • First, the students could have conducted **in-depth interviews with a smaller selection of respondents** as a follow-up to the survey. Unlike the questionnaire, interviews would allow the students to **probe deeper into the reasons** behind respondents' attitudes and behaviours towards e-waste, capturing nuances and personal experiences that fixed survey options cannot. o For example, a respondent who indicated they do not recycle e-waste could be asked to **elaborate on the specific barriers** they face, yielding **richer qualitative data**. • Second, they could have gathered **observational data** by **visiting e-waste recycling points in Ang Mo Kio**, such as the e-waste bins found in community centres or electronics stores, to observe and record how frequently they are used and by which age groups. o This would **provide behavioural data that complements the self-reported data from the survey**, since what people say they do and what they actually do can differ. • Third, they could have collected **secondary data from relevant authorities such as the National Environment Agency (NEA)**, which publishes data on **e-waste collection rates** and **recycling statistics** in Singapore. **AO2**	**5**

Question	Answer	Marks
1 (h)	**With reference to all the resources, evaluate the validity of the students' fieldwork.** Possible Approaches: Candidates could evaluate the validity of the students' fieldwork by **reviewing the hypothesis and choice of research area / location as well as its demographics** shown in Resources 1A, 1B and 2 and **th instrument/questionnaire used to collect the data and the data collected** shown in Resources 3 and 4. The study had a **clear focus and hypothesis**. For the data collection method, candidates **could review the questionnaire** to assess the impact on **accuracy and reliability of data collected** and discuss the degree to which the results shown in Resource 4 supported the hypothesis. Evaluation could also look into the **sample size** and **whether the sample selected is representative** of the adult population in Ang Mo Kio (besides considering age, gender and ethnicity segments, education levels could be considered too) etc. Candidates could consider the **design of the questionnaire,** such as its limitations (order and logic of questions) and how it would have been **augmented if other relevant data such as in-depth interviews and observational data was collected,** noting that such information was vital in investigating whether people are more likely to participate in sustainable practices with respect to e-waste. Levels marked using modified Generic Level Descriptors for H2 fieldwork evaluative questions. **AO3**	**10**

Section B

Cluster 3: Sustainable Future and Climate Change

Question	Answer	Marks
2	Resource 5 is an infographic showing Japan's population data. Resource 6 shows Japan's population by age group in 2022. Resource 7 shows the estimated rate of increase in elderly single-person households by prefecture in Japan in 2050 as compared to 2020. Resource 8 shows the cost of living index comparison by prefecture in Japan in 2024.	
2 (a)	**Using Resources 5 and 6, describe the population characteristics of Japan.** *Award 1 mark for each description.* *Award 1 additional mark for further development of each description, where applicable.* *Award a maximum of 3 marks for description of <u>either</u> Resource 5 <u>or</u> Resource 6.* Possible responses include: • From Resource 5, Japan had a **total population of 124.77 million in 2022**. • Globally, it had the highest percentage of **people aged 65 or older**, i.e. approximately **one-third of its population / 29%** (to be exact) in 2022. • This proportion is expected to increase and **exceed 35% by 2040**. • The proportion of young people **aged 0 – 14** is very small **at 11.6%**, which is less than half of those aged 65 and above. • From Resource 6, there are **slightly more males than females for people aged 0 – 14** (7.6 million vs 7.2 million) **and working age of 15 – 64** (38 million vs 36 million). • However, there are **significantly more female than male seniors aged 65 and older**, i.e. ~20 million female seniors out of 64 million females vs. ~16 million male seniors out of 61 million males / ~31% to 32% of the female population vs ~26% of the male population. ○ **Among seniors**, those **aged 70 – 74 make up the largest group** for both females and males. **AO2**	**5**
2 (b)	**Using Resource 7, describe how the estimated rate of increase in elderly single-person households varies across Japan.** *Award 1 mark for each description.* *Award 1 additional mark for further development of each description, where applicable.* Possible responses include: • The estimated rate of increase in elderly single-person households in Japan is **expected to be 20% or greater in most**	**4**

	of the prefectures / **more than three-quarters of the prefectures** in 2050 as compared to 2020. • The **central and east regions** of Japan would have more elderly single-person households, ranging from **40% to 80%**. • The **north and south regions** would generally have **lower** rates of increase at 20% to 40% in the north and 0% to 20% in the south. • The **anomalies** are two prefectures, one in the central part and another an island in the southernmost part, which show the highest estimated rate of increase at **80% or more**. **AO2**	
2 (c)	**With reference to Resources 5, 6, 7 and 8, suggest possible issues faced by the elderly living in urban areas of Japan.** *Award 1 mark for each suggestion.* *Award 1 additional mark for further development of each suggestion, where applicable.* Possible responses include: • As Japan's elderly population increases (Resources 5 & 6), the **demand for healthcare and caregiving services** also increases. This places significant pressure on hospitals, clinics and eldercare facilities, as they must accommodate a larger number of patients with chronic and age-related conditions. o The shortage of trained caregivers may result in **longer waiting times** for care services and **reduced quality or frequency of personalised attention** for elderly patients. • The rise in elderly single-person households in Japan (Resource 7) can lead to **increased loneliness and social isolation and weaker social support networks**, as many seniors live alone after their children move away or spouses pass on. • Retirees on fixed pensions or limited savings may **struggle to keep up with rising living costs**, particularly in expensive urban areas such as Tokyo. o Resource 8 shows that the **overall cost of living index** is the **highest for Tokyo at 104** (average of 100 for Japan), with **housing at 127.2** which is significantly more expensive than other prefectures. o As housing and healthcare costs increase, seniors facing financial strain may have to **downsize their housing, rely more heavily on government subsidies** or **cut back on food, healthcare and social participation**, which can further **affect** their physical and mental well-being, and hence their **quality of life**. **AO2**	**6**

2 (d)	**Suggest possible strategies to address the issues faced by the elderly living in urban areas of Japan.** *Award 1 mark for each suggestion.* *Award 1 additional mark for further development of each suggestion, where applicable.* Possible responses include: • To manage the growing demand for **healthcare and caregiving services**, Japan could invest in **training and recruiting more caregivers**, including **attracting foreign healthcare workers** from countries like the Philippines and Vietnam. • **Robotics and assistive technology** – areas where Japan already has considerable expertise – could be deployed more widely in eldercare facilities to **supplement human caregivers**, helping with tasks like mobility assistance and medication reminders. • To address the rise in elderly single-person households, community or neighbourhood centres could be strengthened to provide **regular social programmes, meals and check-ins for seniors living alone**. • **Intergenerational housing schemes** – where younger residents share housing with elderly individuals in exchange for reduced rent – could **foster daily social interaction and mutual support**. • Local governments could also establish **volunteer befriending networks** and **emergency alert systems** that allow **seniors to signal for help** quickly. • To help retirees cope with rising living costs, the government could **expand means-tested housing subsidies** and **develop more affordable public housing options for seniors**, particularly in high-cost cities like Tokyo. **AO1**	**6**
2 (e)	**Explain <u>two</u> social factors that influence the liveability of a city.** *Award 1 mark for each explanation.* *Award 1 additional mark for further development of each explanation, where applicable.* *Award maximum of 2 marks if only one social factor is explained.* Possible responses include: • The first social factor is the **availability, affordability and quality of healthcare services**. Cities with well-equipped hospitals, accessible clinics, sufficient medical personnel and affordable healthcare tend to be more liveable, as residents can **receive timely and effective treatment when needed**. ○ Where healthcare infrastructure is **overstretched or underfunded, longer waiting times and reduced quality of care** can diminish residents' well-being and	**4**

	sense of security, lowering the overall liveability of the city. • The second social factor is **education and access to quality schools**. Cities with **well-resourced schools, experienced teachers and good educational outcomes** tend to attract families and are generally considered more liveable. o Education also has **long-term effects on social mobility and economic opportunity**, meaning that cities with strong educational provision tend to foster more stable and cohesive communities over time. **AO1**	
2 (f)	**Explain how climate change might impact the elderly.** *Award 1 mark for each explanation.* *Award 1 additional mark for further development of each explanation, where applicable.* Possible responses include: • One of the most **direct impacts** is through **extreme heat events**, which are becoming **more frequent and intense as global temperatures rise**. Older adults are particularly **susceptible to heat-related illnesses such as heat stroke and dehydration** because the ageing body is less efficient at regulating temperature and detecting thirst. o Those with **pre-existing chronic conditions** such as cardiovascular or respiratory disease face even **greater risk**, and fatalities among the elderly during heatwaves highlight how **deadly** these events can be. • Climate change also increases the **frequency and severity of extreme weather events** such as floods, storms, and wildfires. The elderly may find it **harder to evacuate quickly due to reduced mobility or cognitive decline**, making them more vulnerable to injury or death during such disasters. Those **living alone are at particular risk**, as they may not receive timely warnings or assistance. o **Recovery after such events can also be more difficult** for older adults, both physically and psychologically, as displacement and loss of familiar surroundings can contribute to anxiety and depression. • Air quality is another concern, as rising temperatures and shifting weather patterns driven by climate change directly **worsen air pollution**, e.g. formation of ground-level ozone and **extend pollen seasons**, leading to higher allergen concentrations. Elderly individuals with **respiratory conditions** are especially sensitive to poor air quality, which **can trigger serious health episodes and increase hospitalisation rates**. **AO1**	**5**

Section C

Cluster 3: Sustainable Future and Climate Change

Question	Answer	Marks
3	**'Multiple deprivations faced by slum dwellers living in the city pose the most significant obstacle to progress towards sustainable urban development.'** **Evaluate this statement**. Possible Approaches: Candidates could approach the question by making a judgement on **whether the multiple deprivations faced by slum dwellers** – such as lack of access to clean water, sanitation, secure tenure, healthcare, education and economic opportunities – **pose the most significant obstacle to sustainable urban development (SUD), compared to other obstacles/hindrances such as ineffective waste management and ineffective urban reimaging**. Candidates could argue that **ineffective slum management is a governance failure** that amplifies the multiple deprivations faced by slum dwellers, which pose a significant obstacle/hindrance to SUD because they **undermine all three dimensions of sustainability – social, economic and environmental**. Candidates could **evaluate how the three obstacles/hindrances interact with and reinforce one another**. For instance, ineffective waste management worsens living conditions in slums, deepening deprivations, while ineffective slum management prevents the structural changes needed to address both waste management and deprivations simultaneously. Candidates could also explore the **tension between short-term urban reimaging goals – such as improving a city's global image – and the long-term investments needed to address slum deprivations and achieve SUD**. The complementarity of addressing all three hindrances together could also be examined, with candidates arguing that **progress on one front is unlikely to be sustained without corresponding progress on the others**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	**20**

Question	Answer	Marks
4	**'The most important actor in implementing climate change responses is the state.'** **Evaluate this statement.** Possible Approaches: Candidates could approach the question by making a judgement on whether the most important actor in implementing climate change responses is the **state**, compared to **other actors such as intergovernmental organisations (IGOs), non-governmental organisations (NGOs) and firms.** Candidates should demonstrate an understanding that **climate change is a multi-scalar problem requiring responses at the local, national and global levels**, and that **different actors operate at different scales with varying degrees of influence, resources and accountability**. Candidates could argue that **the most important actor** is the state because states possess the **legislative authority, financial resources and coercive power** needed to enact and enforce meaningful climate policy at the national scale through tools such as carbon taxes, emissions trading schemes and renewable energy investment. As the primary signatories to international agreements such as the Paris Agreement, **states also set the global framework within which all other actors operate**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	**20**